# KARNATAKA RESERVOIR OBSERVATORY

# Bhadra Reservoir

Generated 2026-10-22 · historical monitoring report

## Routine monitoring

Continue collecting field and satellite observations. Available spectral indicators alone do not confirm a harmful bloom; chlorophyll guidance is inactive without dominance evidence.

| Indicator | Available result |
|---|---|
| Satellite observation | 2026-10-12T05:17:39.024000Z |
| Chlorophyll-a | Unavailable - no calibrated concentration |
| Cyanobacteria severity | Class 1 (research model estimate) |
| Forecast | No validated seven-day forecast |
| Valid water pixels | 25065 |

## Cyanobacteria severity classes

Density boundaries describe field-label classes, not universal health-safety thresholds.

Class 1 · <20,000 cells/mL
Source: Field-measured cyanobacteria labels — DrivenData Tick Tick Bloom.

Class 2 · 20,000 – <100,000 cells/mL
Source: Field-measured cyanobacteria labels — DrivenData Tick Tick Bloom.

Class 3 · 100,000 – <1,000,000 cells/mL
Source: Field-measured cyanobacteria labels — DrivenData Tick Tick Bloom.

Class 4 · 1,000,000 – <10,000,000 cells/mL
Source: Field-measured cyanobacteria labels — DrivenData Tick Tick Bloom.

Class 5 · ≥10,000,000 cells/mL
Source: Field-measured cyanobacteria labels — DrivenData Tick Tick Bloom.

## Chlorophyll-a monitoring context

Historical WHO 1999/2003 guidance summarized by EPA. Inactive without independent cyanobacteria dominance evidence. These are not Karnataka regulatory limits.

10 µg/L · WHO historical lower guidance level (1999/2003)
Source: WHO/EPA published monitoring guidance; applicable when cyanobacteria dominance is established.

50 µg/L · WHO historical higher guidance level (1999/2003)
Source: WHO/EPA published monitoring guidance; applicable when cyanobacteria dominance is established.

## Monthly observation coverage

2026-10: 1 accepted acquisition windows / 1 queried windows; 0 missing windows. Chlorophyll mean: unavailable.

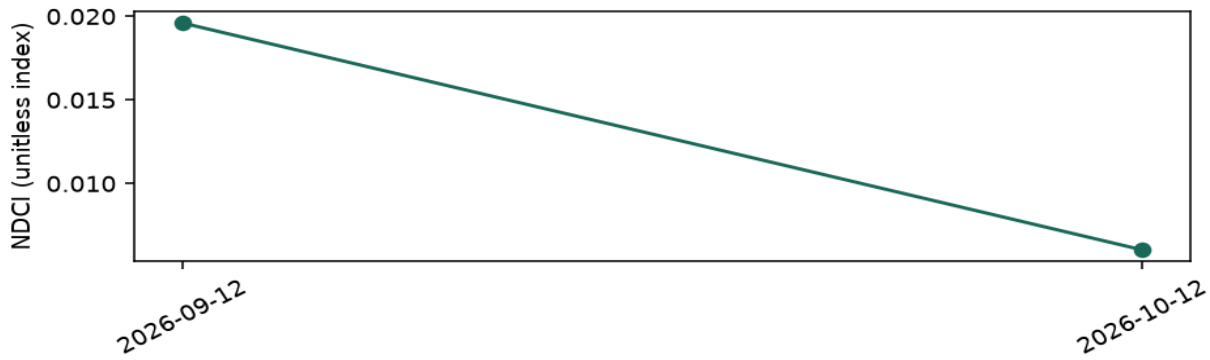
Weather: mean air temperature 24.4 °C; screened rainfall 787.92 mm over 13 accepted dates, with 2 rainfall dates flagged for source review.

2026-09: 1 accepted acquisition windows / 1 queried windows; 0 missing windows. Chlorophyll mean: unavailable.

Weather: mean air temperature 23.6 °C; screened rainfall 732.96 mm over 26 accepted dates, with 4 rainfall dates flagged for source review.

2026-08: 0 accepted acquisition windows / 1 queried windows; 1 missing windows. Chlorophyll mean: unavailable.

Weather: mean air temperature 23.1 °C; screened rainfall 2514.72 mm over 22 accepted dates, with 9 rainfall dates flagged for source review.




## Genuine held-out model evaluation

CAML acquisition pilot: 135 aligned samples; 70 training, 24 validation and 41 test samples in a new geographic split. The original competition split is not used. This is not Karnataka validation.

| Model | Accuracy | Macro F1 | Precision | Recall |
|---|---|---|---|---|
| Neural network | 26.8% | 0.205 | 0.156 | 0.307 |
| Random Forest | 26.8% | 0.266 | 0.292 | 0.313 |

Held-out performance is weak. These exploratory estimates do not drive monitoring alerts. The acquisition cohort is class-enriched and quality-selected. Scores do not estimate full-CAML population performance. Supplied categorical severity labels are used because the raw abundance unit declaration is inconsistent. No corrected density is inferred.

Dataset: https://doi.org/10.5067/SeaBASS/CAML/DATA001

## Methods, provenance and limitations

Sentinel-2 L2A pixels are aligned to a 60 m grid. Mapped-water polygons, SCL water classification, cloud/shadow exclusion, a 60 m shoreline buffer, and valid-reflectance filtering are applied. Minimum counts and coverage are project screening choices. No missing measurements are synthesized.

Training labels, when supplied, are field observations collected in the United States. Held-out evaluation there is not field validation in Karnataka. NDCI/NDWI/FAI are spectral indicators, not toxin concentrations. Weather is gridded NASA POWER meteorology, not lake-water temperature. Provider rainfall at or above the project sanity limit of 200 mm/day is retained for review and excluded from monthly totals.

Sources: https://www.drivendata.org/competitions/143/tick-tick-bloom/page/650/

Monitoring guidance: https://www.epa.gov/habs/world-health-organization-who-1999-guideline-values-cyanobacteria-freshwater

Satellite source: https://planetarycomputer.microsoft.com/api/stac/v1/collections/sentinel-2-l2a/items/S2B_MSIL2A_20261012T051739_R062_T43PER_20261012T091927

Weather source: https://power.larc.nasa.gov/api/temporal/daily/point?parameters=T2M%2CPRECTOTCORR%2CWS2M&community=AG&latitude=13.6145573&longitude=75.5671397&start=20260717&end=20261015&format=JSON